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PROJECT AND TEAM INFORMATION

Project Title

SlopeShield - Intelligent Landslide Risk and Safe Route System

Team Name: *Team #*	*The Debug Divas*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

SlopeShield is a landslide risk assessment and route planning system designed for Uttarakhand's Yatra routes. Heavy rainfall often triggers landslides in hilly areas which results in blocking roads and putting travellers and rescue teams at risk. While traditional hazard maps may not reflect changing conditions, most navigation apps focus on finding the shortest route. SlopeShield aims to address this gap by assessing 11 important locations using rainfall, forecast and humidity data from the OpenWeatherMap API. Each location receives a risk score from 0 to 100 and is classified as Low, Moderate, High or Critical. The system then uses these scores to evaluate 14 road corridors and applies Dijkstra's algorithm with a custom min-heap priority queue to find a suitable lower-risk route. The results are displayed through a REST backend and an interactive map built with Leaflet.

Updated Project Approach and Architecture (2 pts)

SlopeShield is developed using Python and C++ and follows an object-oriented approach. The system is organised into four main layers which are data management, risk assessment, route planning and presentation. The data layer stores location and road information in CSV and JSON files. The risk assessment layer uses weather information to calculate risk levels. It supports mock data when live weather data is unavailable. The route planning layer uses a graph,a priority queue, adjacency lists and Dijkstra's algorithm to recommend routes based on road risk. The presentation layer uses Leaflet and OpenStreetMap to display risk zones,locations and routes on an interactive map.

The SlopeShieldBackend class connects the different components of the system. The frontend and backend exchange data through REST APIs using JSON while the weather information is retrieved through the OpenWeatherMap API. The project uses Python libraries, C++ STL, HTML,JavaScript, CSS, Leaflet and OpenStreetMap, with Git and GitHub for version control. This structure makes the system easier to manage, test and improve in the future.

Tasks Completed (7 pts)

Task Completed	Team Member
Dataset, risk formula and risk classification. weather fetching. risk test cases.	Smriti Singh
Web interface: LeafLet map, route planner, corridors table and hazard simulator.	Sanya Verma
Road Network graph, min-heap priority queue, Dijsktra's routing and route test cases.	Jahnvi Verma
OOP architecture, CSV/JSON handlers, backend facade, REST API and integration tests.	Anshika Gaur

Challenges/Roadblocks (7 pts)

1. Weather API: API authentication issues required a mock data mode for testing without live weather accessing.
2. Limited Data: Detailed terrain data was not available, so representative values are used for better data planning
3. Risk Calculation: Selecting suitable risk weights and thresholds required testing different scenarios.
4. Route Planning:Finding a suitable route that considers both road distance and landslide risk involved testing different hazard scenarios.
5. Compatibility:A local server and portable C++ code were used to make sure that the project runs smoothly on different systems.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Real slope and elevation data.Testing system using real time data to check its accuracy.	Smriti Singh
Road geometry on the map, final demo and UI polish	Sanya Verma
Support for one way roads and extra route tests.	Jahnvi Verma
Per session simulation. GitHub release.API hardening.	Anshika Gaur
Final documentation and report	All members

Project Outcome/Deliverables (2 pts)

The main outcome of SlopeShield is a system that assesses landslide risk at 11 locations in Uttarakhand using weather conditions. It assigns each location with a risk score between 0 and 100 and classifies the risk level accordingly. The system also suggests the shortest and lower risk routes between different locations. It's hazard simulator lets users explore different landslide situations and see how they may affect the available routes.
The project includes Python and C++ modules for risk calculation, a road network that uses a custom min-heap and Dijkstra's algorithm for route planning and an OOP based backend with CSV/JSON file handling and REST APIs. It features an interactive Leaflet dashboard for viewing routes and testing hazard scenarios. Other deliverables also include location, road, weather and route data, test cases, project documentation, member guides, a README file and final demonstration.

Progress Overview (2 pts)

The core SlopeShield system has been developed and is functional, route planning, including risk scoring, backend APIs, automated testing and the web dashboard . Approximately 80 percent of the project has been completed and the current automated test suites are passing.
The project has made steady progress with additional features such as the C++ implementation, hazard simulator and mock-weather support. However integrating detailed terrain data,validating the system with live weather data, improving road mapping accuracy, and completing the final documentation are still in progress. These tasks will be addressed during the remaining phase of the project.

Codebase Information (2 pts)

GitHub Repository: https://github.com/SmritiSingh118/SlopeShield
The project consists of core modules for OOP models,risk scoring, data handling and route planning along with backend services, weather and route scripts, a C++ implementation and a Leaflet based frontend. The project also includes datasets,test cases, configuration files and a main Python file that runs the application. The major development milestones included preparing the dataset and risk formula, adding weather support with a Sample data backup, implementing the graph, min-heap priority queue and Dijkstra's algorithm, organizing the code using OOP concepts integrating the REST API with the frontend, developing the hazard simulator and implementing the system in C++.

Testing and Validation Status (2 pts)

Test Type	**Status (Pass/Fail)**	**Notes**
Risk scoring	Pass	Dry winter site remains moderate. cloudburst at Rudraprayag gives critical (77.1). Heavy rain at Mussoorie gives high.
Route engine	Pass	Shortest and risk-aware routes match expected paths in dry, hazard and monsoon scenarios.
Integration	Pass	OOP models, JSON/CSV handlers, backend pipeline, hazard detour routing and REST API tested.
Live OpenWeatherMap test	Pending	Currently validated with mock data. Requires an active API key for live testing.

Deliverables Progress (2 pts)

Datasets (locations, roads, weather and scores)
Risk scoring module(python and c++)
Road Graph, Dijkstra routing and min heap
OOP backend, file handling and REST API
Leaflet web dashboard and hazard simulator
Test suite
Real DEM terrain data and live weather validation
Project documentation and final report
Final documetation